

DSAA3073: Theories in Data Science

Week 5 Open Investigation

Stabilizing a Spectroscopic Inverse Problem

Ridge regression beyond spherical fixed-design noise

Basic analysis. A portable spectrometer with an unstable calibration direction

Use only the bias-variance, spectral-conditioning, Tikhonov, and ridge results from the Week 5 Learning Sheet

A food-safety laboratory uses a portable near-infrared spectrometer to estimate two latent chemical components from a sample. After centering the calibration data, the design Gram matrix has eigendecomposition

$$X^T X = Q \operatorname{diag}(100, 1) Q^T.$$

The first direction is strongly determined by the calibration sample; the second is a weak spectral direction. For the current batch, the transformed response vector is

$$Q^T X^T y = (100, 2)^T.$$

The laboratory must decide whether to release the unregularized estimate or use ridge with $\lambda = 4$. Excessive variation in the second component can trigger a false product recall, whereas excessive shrinkage can conceal a genuine chemical signal.

For the theoretical comparison, use the provisional fixed-design model

$$y = X\beta^* + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0, \quad \operatorname{Cov}(\varepsilon \mid X) = \sigma^2 I_n, \quad Q^T \beta^* = (1, 1)^T.$$

For the numerical risk comparison below, take $\sigma^2 = 1$.

Basic Problem 1. Compute and interpret the two candidate estimates

Derive the OLS coefficient and the ridge coefficient at $\lambda = 4$ in the Gram-eigenvector basis. Compute the directional shrinkage factors. Relate the numerical change to the singular conditioning of the calibration problem, and state which part of the estimate is most sensitive to measurement noise.

Basic analysis. Evaluate the statistical price of stabilization

A deterministic inverse calculation becomes a risk calculation only after a probability model is declared

Basic Problem 2. Complete the fixed-design bias-variance diagnosis and make a recommendation

Derive the directional mean, bias, and variance of ridge under the provisional model. Compare the total coefficient mean-squared error for $\lambda = 0$ and $\lambda = 4$. Then give the laboratory a conditional recommendation. Your conclusion must distinguish uniqueness and numerical stability from predictive or coefficient risk.

Thinking investigation. Formalize the assumption-guarantee contract

Keep the estimator visible while changing one component of the data-generating structure

Represent the analysis by

$$(\mathcal{A}, \mathcal{P}, \hat{\beta}, \mathcal{G}),$$

where $\mathcal{A}$ is the sampling model, $\mathcal{P}$ the penalty geometry, $\hat{\beta}$ the estimator, and $\mathcal{G}$ the moment or risk statement. The Basic model uses fixed X , spherical conditional noise, and $\mathcal{P}(\beta) = \|\beta\|^2$.

Throughout the Thinking investigation, write $G := X^T X$ for the design Gram matrix.

Investigation 1. Replace spherical noise by a general covariance

Suppose $\mathbb{E}[\varepsilon \mid X] = 0$ remains valid but $\text{Cov}(\varepsilon \mid X) = \Sigma$, where Σ is positive semidefinite and need not be a multiple of the identity. Determine which part of the Week 5 ridge moment theorem survives. Derive the exact covariance of $\hat{\beta}_\lambda$. In the Q basis, explain why ridge directions need no longer be statistically independent even though Q diagonalizes $X^T X$.

For $\lambda = 4$, specialize to

$$B := Q^T X^T \Sigma X Q = \begin{pmatrix} 100 & 10 \\ 10 & 4 \end{pmatrix}.$$

Calculate the resulting covariance matrix in the Q basis.

Thinking investigation. Change the penalty geometry

A positive tuning parameter does not by itself guarantee a unique minimizer

Investigation 2. Replace the Euclidean penalty by a quadratic scientific prior

Let Γ be positive semidefinite and replace $\|\beta\|^2$ by $\beta^T \Gamma \beta$. Derive the normal equations and give a necessary and sufficient condition for a unique minimizer. If $G = X^T X$ and Γ share eigenvectors with eigenvalues μ_j and g_j , derive the directional filter and describe exactly what happens when $\mu_j = g_j = 0$.

Thinking investigation. Let the calibration design be random

Conditional validity does not imply the same unconditional formula

Investigation 3. Move from a fixed design to repeated random calibration samples

Suppose X is random but, conditional on X , the correctly specified model and spherical noise assumptions remain valid. Decide whether the Week 5 theorem is lost or merely conditional. Derive unconditional expressions for $\mathbb{E}[\hat{\beta}_\lambda]$ and $\text{Cov}(\hat{\beta}_\lambda)$ using iterated expectation and total covariance. Identify the additional source of variability that is absent from the fixed-design calculation.

Synthesis. State the general Week 5 conclusion

Return to the spectrometer case. State which conclusion is supported under each of the three modified assumptions, and propose one additional calibration measurement that would distinguish weak spectral information from correlated measurement noise.